

IT 362 Course Project
Data Science
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Sentiment analysis of opinions about flu vaccines

Project Group #3

Student Name:	KSU ID:
Jood Abdullah Alkhneen	445201348
Tala Abdullah Alqahtani	445204552
Shatha Hamad mana	445202291
Rana Abdulrhman Alsalman	445201204
Lama Abdullah Almubarak	445200338

Supervised by:
Dr. Lama Al-Sudais

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Introduction

Influenza vaccination is widely recognized as one of the most effective measures for preventing seasonal influenza and reducing its associated health problems. Despite its proven benefits, public attitudes toward flu vaccines remain diverse. While many individuals support vaccination and view it as essential for protecting personal and public health, others express concerns regarding vaccine safety, side effects, effectiveness, and trust in health authorities.

The widespread use of online platforms has enabled people to openly share their experiences, beliefs, and opinions about flu vaccination. These digital discussions generate large volumes of unstructured textual data that offer important findings into public perception and sentiment. Analyzing such data provides an opportunity to better understand vaccine acceptance, hesitancy, and the factors influencing public trust.

This project focuses on applying sentiment analysis techniques to examine public opinions about flu vaccines using data collected from multiple online sources. By analyzing sentiment patterns and common themes, this study aims to better understand how flu vaccines are perceived across different information sources.

Research Question:

What is the overall sentiment and thematic framing of flu vaccination across social media, regulatory reports, and news media sources?

Literature Review

Study 1: Examining the Negative Sentiments Related to Influenza Vaccination

Problem addressed:

This study investigated negative public sentiments toward influenza vaccination, focusing on vaccine hesitancy driven by misinformation and distrust in vaccination policies.

Dataset used:

Unstructured textual data from Twitter, consisting of 261,613 English tweets related to influenza vaccination collected between 2017 and 2022.

Methods/models applied:

BERT-based deep learning models were used for sentiment classification, and BERTopic was applied for topic modeling to identify major themes within negative sentiments.

Key results:

Negative sentiment toward influenza vaccination increased significantly after 2020. The dominant concerns involved distrust in government policies and the spread of misinformation.

Study 2: Examining Public Messaging on Influenza Vaccine over Social Media

Problem addressed:

This study analyzed how organizations communicate about influenza vaccination on social media and assessed the scope and effectiveness of public health messaging.

Dataset used:

235,261 English tweets posted by organizational accounts between 2017 and 2023, collected using the Twitter API.

Methods/models applied:

Unsupervised deep learning topic modeling using BERTopic.

Key results:

Four main topics emerged: vaccination campaigns, vaccination during pregnancy, age recommendations, and vaccine importance during pregnancy. The study found limited diversity in public messaging content.

Study 3: Examining Public Sentiments and Attitudes Toward Vaccination

Problem addressed:

This research explored public sentiments, vaccine hesitancy, and attitudes toward vaccination using large-scale social media discussions.

Dataset used:

2,944,530 English tweets collected from individual users between January and April 2021.

Methods/models applied:

CorEx topic modeling combined with VADER sentiment analysis. Preprocessing included duplicate removal and noise filtering.

Key results:

Six main thematic groups were identified, including vaccine policies, side effects, and hesitancy. Positive sentiment was generally higher, but hesitancy-related topics remained strongly negative.

Study 4: Health Decision-Making Preferences and Influenza Vaccination

Problem addressed:

This study examined how individuals' health decision-making styles influence vaccine confidence and hesitancy.

Dataset used:

Survey data from nationally representative U.S. adult samples (1005 participants in 2016 and 1020 in 2018).

Methods/models applied:

OLS regression analysis based on health decision-making preference scales.

Key results:

Trust in science-based medicine and physician recommendations strongly increased vaccine confidence and reduced hesitancy.

Study 5: Public Perceptions of Mandatory Influenza Vaccination Policies

Problem addressed:

This study explored public attitudes toward mandatory influenza vaccination policies for healthcare workers.

Dataset used:

1,163 online comments from Canadian news websites responding to vaccination-related articles.

Methods/models applied:

Qualitative thematic analysis with sentiment classification.

Key results:

Nearly half of commenters expressed negative sentiment toward flu vaccines, and most opposed mandatory vaccination policies due to concerns over personal freedom and vaccine safety.

Comparison of Existing Studies

Common Datasets and Features

Most studies relied on unstructured textual data from online platforms, particularly Twitter. Common features included opinion text, timestamps, and sentiment labels (positive, negative, neutral). The primary research focus across studies was understanding vaccine hesitancy and public attitudes.

Differences in Modelling Approaches

Deep learning models such as BERT and BERTopic were used in Studies 1 and 2. Study 3 combined topic modeling with lexicon-based sentiment analysis (VADER). Study 4 applied statistical regression techniques using structured survey data, while Study 5 used qualitative thematic analysis. These approaches ranged from advanced machine learning to traditional statistical and qualitative methods.

Strengths and Limitations

Social media-based studies benefit from large-scale real-time data but are limited by language constraints and demographic representation. Survey-based research provides behavioral insights but lacks dynamic public discourse. Qualitative studies offer detailed thematic understanding but involve smaller datasets. Overall, most studies focus on sentiment trends but usually analyze only one type of data source.

Aspect	Study 1	Study 2	Study 3	Study 4	Study 5
Research Focus	Negative sentiments toward influenza vaccination	Public messaging about influenza vaccine on social media	Public sentiments and attitudes toward vaccination	Health decision-making preferences affecting vaccine confidence	Public perceptions of mandatory influenza vaccination policies
Dataset Used	Twitter tweets about influenza vaccination	Tweets from organizational social media accounts	Large-scale tweets from individual users	Survey data from U.S. adults	Comments from Canadian news websites
Dataset Size	261,613 tweets (2017–2022)	235,261 tweets (2017–2023)	2,944,530 tweets (Jan–Apr 2021)	1005 participants (2016) and 1020 participants (2018)	1,163 online comments
Data Type	Unstructured textual data	Unstructured social media posts	Unstructured social media discussions	Structured survey data	Textual online comments
Methods / Models Used	BERT for sentiment classification and BERTopic for topic modeling	BERTopic for unsupervised topic modeling	CorEx topic modeling + VADER sentiment analysis	OLS regression analysis	Qualitative thematic analysis with sentiment classification
Main Topics Identified	Vaccine hesitancy and misinformation	Vaccination campaigns, pregnancy vaccination, age recommendations	Vaccine policies, side effects, hesitancy	Health decision-making styles and vaccine confidence	Mandatory vaccination policies and public reactions
Key Findings	Negative sentiment increased after 2020 due to distrust and misinformation	Public messaging topics were limited and focused on specific vaccination themes	Overall sentiment was positive, but hesitancy topics remained strongly negative	Trust in science and physician recommendations increased vaccine confidence	Nearly half of commenters expressed negative attitudes toward mandatory vaccination

Strengths	Large social media dataset capturing public opinion	Focus on official public health communication	Very large dataset capturing diverse discussions	Provides behavioral insights from survey data	Captures detailed opinions from real user discussions
Limitations	Limited to Twitter users and English tweets	Limited diversity in messaging topics	Short time window for data collection	Smaller sample compared to social media datasets	Smaller dataset and qualitative analysis

What Is Missing or Limited in Previous Work

Although existing research provides useful insights into public attitudes toward vaccination, several limitations remain. Many studies rely on single data sources and English-language content, reducing generalizability. Additionally, limited research integrates multiple perspectives, such as regulatory data and media coverage. Some qualitative studies also use relatively small sample sizes, restricting broader conclusions.

How This Project Builds Upon Existing Studies

This project focuses specifically on public opinions toward influenza vaccination and collects data from multiple sources, including online discussions, official health reports, and news media coverage, to provide a broader and more diverse perspective. By emphasizing influenza vaccines and utilizing varied data sources, this study aims to provide a clearer understanding of vaccine acceptance and hesitancy.

Data Sources and Bias Evaluation

Data Source

This project combines data from multiple publicly available APIs to obtain diverse perspectives on public opinions regarding flu vaccination. By integrating social media discussions, regulatory health reports, and news media coverage, the dataset aims to provide a broader understanding of how flu vaccines are perceived across different information channels.

Source 1: YouTube Data API v3

The YouTube Data API was utilized to collect public comments related to flu vaccination. Relevant videos were identified using keyword-based searches such as “flu vaccine,” “flu shot,” and “influenza vaccine,” and associated comment threads were extracted.

- Number of observations: 3,896 comments
- Main features and data types:
 - comment_text ((string) unstructured text)
 - publishedAt (datetime)
 - video_id (string)
- Data format: JSONL (unstructured textual data)

This source captures spontaneous public reactions and discussions surrounding flu vaccines.

Source 2: openFDA API

The openFDA API was employed to retrieve reports related to adverse events following influenza vaccination from the FDA’s pharmacovigilance system.

- Number of observations: approximately 1,000 reports
- Main features and data types:
 - report_id (string)
 - reaction (categorical/text)
 - patient_age (numeric)
 - report_date (datetime)
- Data format: Structured JSON records

This dataset provides an official regulatory perspective on vaccine-related outcomes rather than subjective opinions.

Source 3: GNews API

The GNews API was used to collect news articles discussing influenza vaccination through keyword-based queries.

- Number of observations: 110 articles
- Main features and data types:
 - title (string)
 - description (string)
 - content (string – semi-structured text)
 - publishedAt (datetime)
 - source (string)
- Data format: Semi-structured JSON

This source reflects how flu vaccination is portrayed in media coverage.

Bias Evaluation

Given the integration of multiple data sources, several forms of bias may influence the dataset.

Representation Bias

The collected data may not equally represent all population groups. YouTube comments primarily reflect individuals who actively engage online, potentially overrepresenting younger or more digitally active users. openFDA reports tend to capture individuals who experienced and reported adverse events, which may emphasize negative outcomes. News articles are shaped by editorial priorities and may highlight controversial topics.

Consequently, individuals with limited internet access or those who do not report vaccine experiences may be underrepresented.

Measurement Bias

Bias may arise from the manner in which data is produced and collected. Social media comments may include sarcasm, emotional language, or misinformation. openFDA reports depend on voluntary reporting and the accuracy of medical documentation. News articles reflect journalistic framing rather than neutral observation. In addition, keyword-based API searches may influence which records are retrieved.

Historical Bias

Public discussions about vaccines are often shaped by previous health crises, misinformation campaigns, and societal trust in healthcare institutions. Adverse event reporting systems historically capture more severe cases, while media narratives may reflect longstanding public debates and inequalities in healthcare access.

As a result, the dataset may mirror existing societal attitudes rather than an unbiased distribution of opinions.

Objectives

The main objective of this project is to analyze public opinions about flu vaccines using unstructured and semi-structured textual data collected from multiple online sources.

Specifically, this study aims to:

1. Identify the overall sentiment (positive, negative, and neutral) expressed toward flu vaccines.
2. Examine common concerns and themes associated with negative and positive opinions.
3. Compare sentiment patterns across different data sources, including social media, regulatory reports, and news media.
4. Explore changes in public sentiment over time.
5. Generate insights that can support public health awareness and communication strategies.

Method

This project follows a structured data collection and analysis workflow to examine public sentiment toward flu vaccines using unstructured and semi-structured textual data obtained from multiple online sources.

First, data was collected using publicly available APIs, including the YouTube Data API, openFDA API, and GNews API. Keyword-based queries such as “flu vaccine,” “flu shot,” and “influenza vaccine” were used to retrieve relevant records. The collected data was stored in its raw format (JSON/JSONL) without any modifications to preserve its original structure.

After data collection, preprocessing steps were applied to prepare the textual data for sentiment analysis. These steps are expected to include removing duplicates, handling missing values, text normalization, tokenization, and noise removal.

Following preprocessing, sentiment analysis techniques will be employed to classify the collected text into positive, negative, or neutral categories. Both traditional machine learning approaches and deep learning models may be explored to compare performance.

In addition to sentiment classification, topic modeling methods will be used to identify common themes and concerns expressed in public discussions about flu vaccination. Temporal analysis may also be conducted to examine how sentiment trends change over time.

Finally, the results will be visualized and interpreted to provide insights into public attitudes, potential misinformation patterns, and overall perception of flu vaccines.

Challenges

During Phase 1 of the project, several challenges were encountered in the data collection process. Initially, web scraping techniques were attempted on platforms such as Reddit and health review websites. However, these attempts were blocked by access restrictions and HTTP 403 errors, making automated scraping unreliable for large-scale data collection.

Additionally, some sources provided limited or inconsistent data volumes, which made it difficult to build a scalable and comprehensive dataset. As a result, the team shifted to using API-based data collection methods, which offered more stable and structured access to information.

Despite the advantages of APIs, further challenges arose, including configuring authentication systems (such as Google Cloud setup), handling pagination across large result sets, and managing free-tier rate limits imposed by certain services, particularly the daily request limits of the GNews API. These limitations required careful adjustment of query parameters and strategic selection of data sources to ensure sufficient raw data was collected without modification.

To mitigate these challenges in future stages, it is recommended to apply for extended API access tiers, utilize automated scheduling tools to distribute data requests over time, and explore official research access programs provided by data platforms. Moreover, maintaining multiple alternative data sources can reduce dependency on a single platform and improve overall data availability.

Data Processing and Cleaning

Before performing exploratory data analysis, several preprocessing steps were applied to prepare the datasets for analysis.

These steps included removing duplicate records, handling missing values, and cleaning textual data.

Text preprocessing involved converting all text to lowercase, removing URLs, punctuation, and special characters, and filtering common stopwords to reduce noise in the text.

In addition, cleaned text columns such as `text_clean` and `text_processed` were created to facilitate further analysis such as word frequency analysis and sentiment analysis.

Exploratory Data Analysis (Primary Data)

GNews Dataset

The GNews dataset contains news articles related to flu vaccines collected from various online news sources, it was selected because it helped us understand how vaccines were presented in media coverage. The dataset includes several fields such as the article title, description, article content, publication date, and the source of the article.

This dataset was analyzed using Exploratory Data Analysis (EDA) techniques in order to understand the structure of the data, identify patterns, and explore how flu vaccines are discussed in news media.

Dataset Overview

The dataset contains 110 news articles collected from multiple news websites.

Each article includes several attributes such as the title, description, full content, publication date, and the source of the article.

In addition, processed text columns such as `text_clean` and `text_processed` were created to prepare the data for text analysis tasks in later phases of the project.

Dataset Structure

The GNews dataset contains several important columns describing news articles related to flu vaccines.

Key attributes include:

- **Title:** the title of the news article
- **Description:** a short summary of the news article
- **Content:** the main text or body of the article
- **PublishedAt:** the date and time when the article was published
- **Source:** the news source or website that published the article
- **Url:** the link to the original news article
- **text_clean:** cleaned version of the article text used for analysis
- **text_processed:** processed text prepared for text analysis
- **article_length:** the length of the news article measured by the number of words in the content

These attributes help analyze news content , sources ,and publication patterns related to flu vaccines.

Missing Values Check

The dataset was examined to check for missing values.

The results show that there are no missing values in any of the dataset columns, And this means the dataset is complete and ready for further analysis without additional cleaning.

Exploratory Data Analysis Results

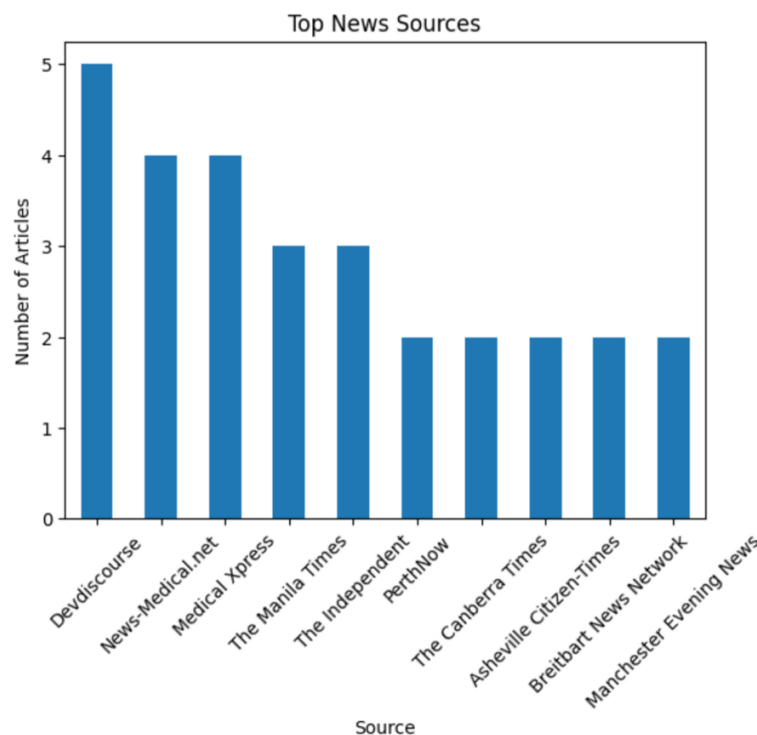


Figure 1: Top news sources publishing articles related to flu vaccines

This figure shows the news sources that published the highest number of articles related to flu vaccines.

Some sources appear more frequently than others , indicating that certain news websites discuss flu vaccines more actively than others.

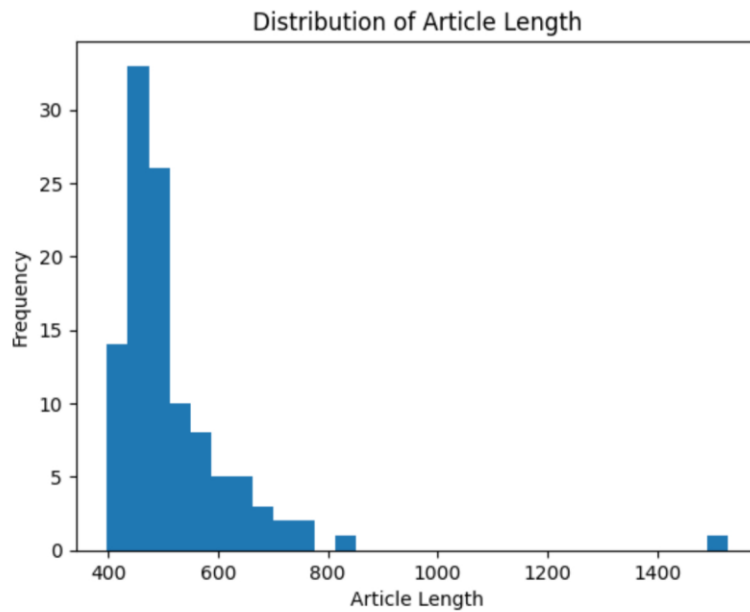


Figure 2: Distribution of article length

This figure shows the distribution of article lengths in the dataset.

Most of the articles have a moderate length , which suggests that news articles discussing flu vaccines typically provide a reasonable amount of information without being excessively long.

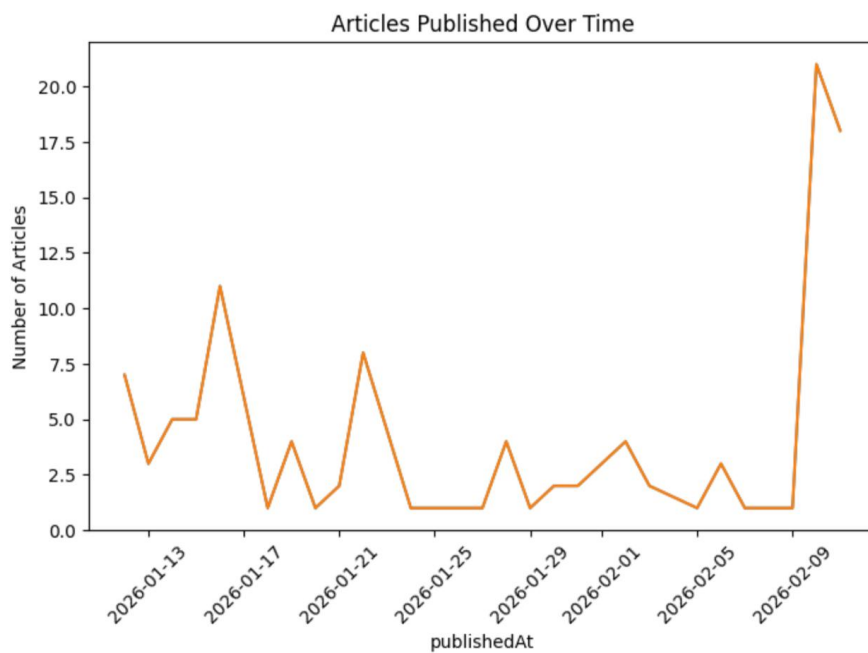


Figure 3: Number of articles published over time

This figure illustrates how the number of news articles related to flu vaccines changes over time.

From the chart , we can see that the number of articles doesn't stay the same and tends to vary across different periods.

The number of articles may increase during periods when flu-related news becomes more important or when new developments about vaccines occur.

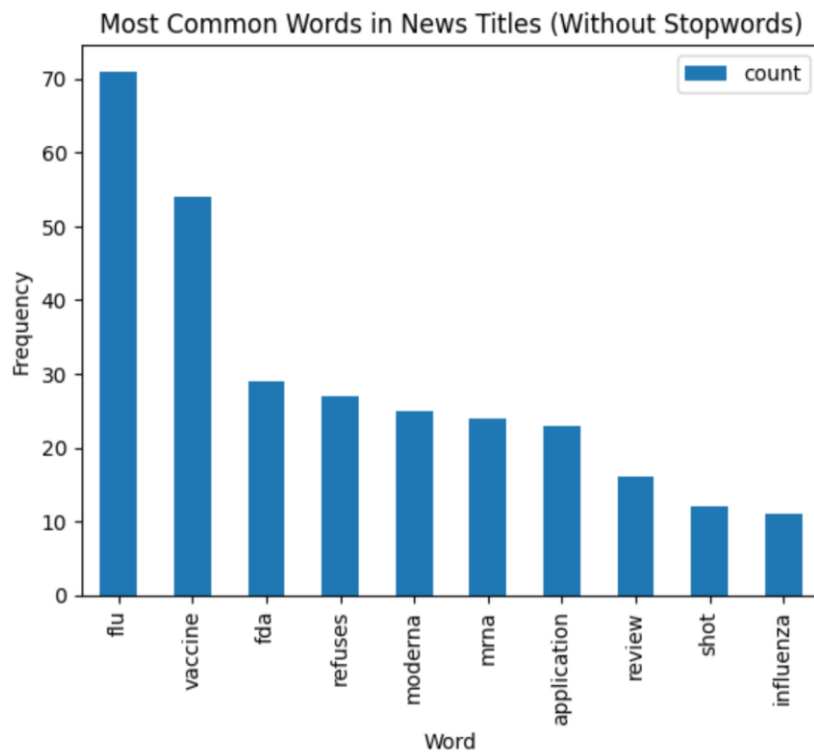


Figure 4: Most common words appearing in the news articles

This chart shows the most frequently used words in the news articles related to flu vaccines.

Words such as "flu", "vaccine", and other related terms appear frequently, which confirms that the dataset focuses on discussions about influenza vaccines.

Summary

The exploratory data analysis helped us understand the structure and characteristics of the GNews dataset.

The dataset includes articles from many different news sources discussing flu vaccines.

The analysis also highlights patterns in article publication and the most frequently used terms related to flu vaccines.

These insights provide a useful foundation for further analysis such as sentiment analysis and topic modeling in the next phase of the project.

OpenFDA Dataset

The OpenFDA dataset contains reports of side effects related to flu vaccines. These reports are collected from healthcare providers, patients, and medical institutions and published through the FDA adverse event reporting system.

The dataset includes several fields such as patient age, gender, reported reactions, and report identifiers. These details help analyze the characteristics of side effects reported after flu vaccination.

Exploratory Data Analysis (EDA) was conducted to understand the structure of the dataset, identify patterns in patient demographics, and examine the most common reactions reported after flu vaccines.

Dataset Overview

The dataset contains vaccine adverse event reports collected from the OpenFDA reporting system.

Each record represents a report describing a reaction experienced after vaccination. The dataset includes demographic information about patients and details about reported symptoms.

The analysis focuses on understanding the distribution of patient demographics and identifying the most common reactions reported after flu vaccination.

Dataset Structure

The OpenFDA dataset contains information about adverse event reports related to flu vaccines.

Key attributes include:

- **report_id:** unique identifier for each adverse event report
- **patient_age:** the age of the patient reported in the dataset
- **age_group:** categorized age group of the patient
- **patient_sex:** gender of the patient
- **Reaction:** the medical reaction or symptom reported after vaccination
- **report_date:** the date when the adverse event report was submitted
- **text_clean:** cleaned version of the text data used for analysis
- **text_processed:** processed text prepared for further analysis

These attributes allow analysis of patient demographics and commonly reported vaccine reactions.

Missing Values Check

The dataset was checked to find missing values in different columns. The results show that some demographic variables have missing values , especially patient_age and age_group. However , the main report information such as report identifiers and reaction descriptions is still available for analysis.

Even though some demographic data is missing ,the dataset still provides enough information to analyze reaction patterns and patient characteristics.

Exploratory Data Analysis Results

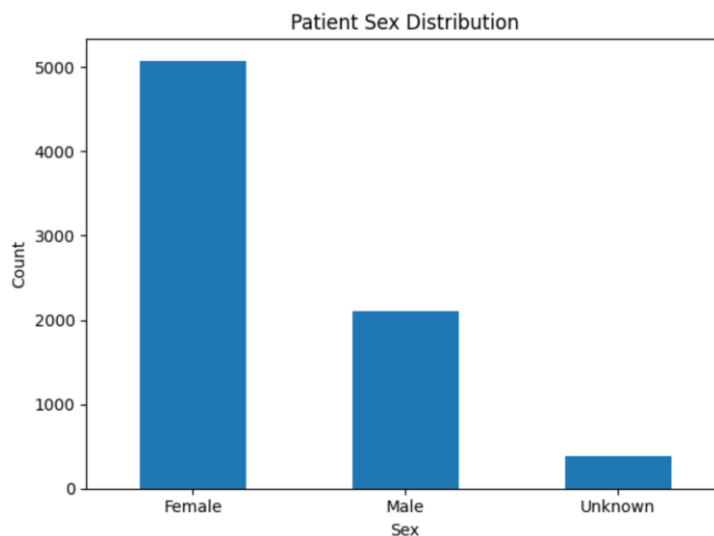


Figure 5: Distribution of patient gender in reported vaccine reactions

This figure shows the distribution of reported vaccine reactions by patient gender.

The results indicate that female patients appear more frequently in the reports than male patients, while a smaller portion of the reports have an unknown gender value. This difference may reflect variations in reporting behavior or differences in healthcare reporting patterns.

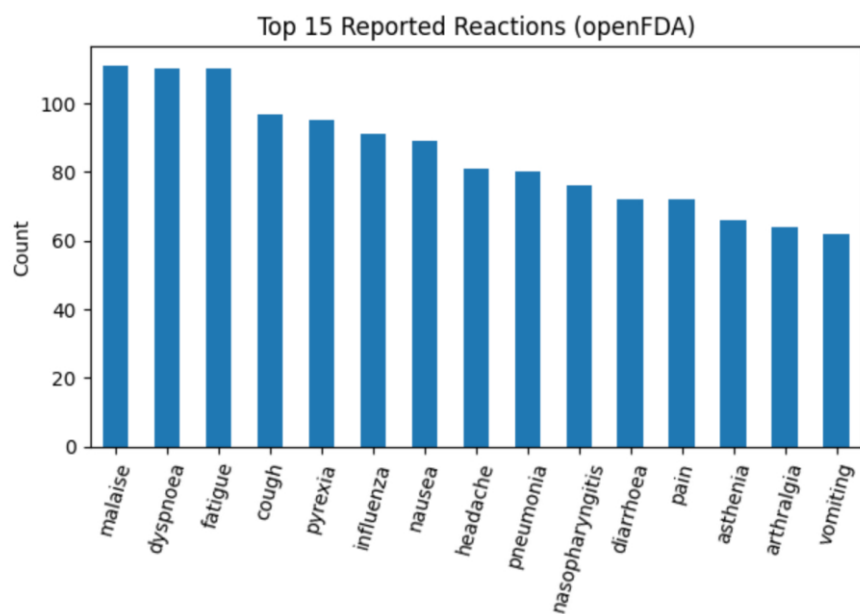


Figure 6: Most common reported reactions related to flu vaccines

This figure shows the most frequently reported reactions related to flu vaccines. Symptoms such as malaise, dyspnoea, fatigue, cough, and pyrexia appear among the most commonly reported reactions.

These symptoms are usually related to the body's immune response after vaccination and are commonly reported in vaccine safety monitoring systems.

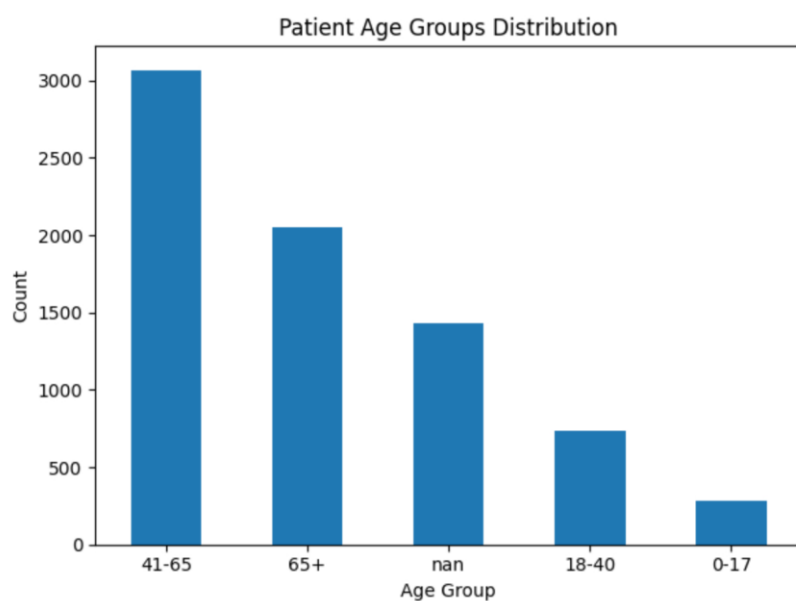


Figure 7: Distribution of patient age groups in the dataset

This figure shows the distribution of reported cases across different patient age groups.

The chart indicates that most reports are associated with individuals in the 41–65 age group, followed by patients aged 65 and above. Fewer reports are observed among younger individuals.

This may suggest that adverse event reports are more commonly documented among middle-aged and older populations.

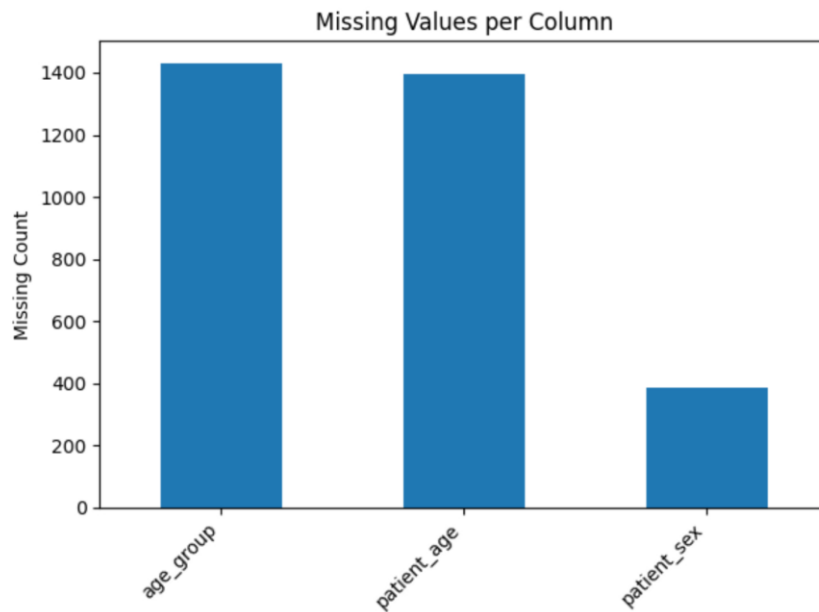


Figure 8: Missing values across dataset columns

This figure presents the number of missing values in several demographic variables.

The chart shows that age_group and patient_age contain the largest number of missing values, while patient_sex contains fewer missing entries. This indicates that some demographic information was not provided in certain reports, which is common in large healthcare reporting datasets.

Summary

The exploratory data analysis helped understand the structure and characteristics of the OpenFDA dataset.

The analysis revealed demographic patterns in reported vaccine reactions, including differences in gender distribution and age groups of reported cases. Additionally, the most common symptoms associated with flu vaccine reactions were identified.

Although some demographic attributes contain missing values, the dataset still provides major insights into vaccine safety reports.

These findings provide an important foundation for further analysis in the next phase of the project, such as text analysis and predictive modeling.

YouTube Dataset

The YouTube dataset contains comments collected from videos discussing flu vaccines. These comments reflect public opinions , reactions , and discussions related to vaccination.

The dataset includes textual comments posted by users as well as metadata such as comment identifiers , video identifiers ,timestamps, and the number of likes received by each comment.

Exploratory Data Analysis (EDA) was performed to better understand the dataset structure, examine patterns in comment activity over time, analyze commonly used words in discussions , explore sentiment distribution, and study the characteristics of comment lengths.

Dataset Overview

After preprocessing and removing duplicates, the dataset contains 3,781 comments collected from YouTube videos discussing flu vaccines.

Each row represents a single comment written by a user. The dataset includes both textual information and metadata related to the comment.

This dataset is useful for analyzing public discussions on social media and identifying common concerns, opinions, and reactions toward flu vaccines.

Dataset Structure

The dataset contains several important columns that describe each comment.

Key attributes include:

- **CommentId:** unique identifier for each comment
- **VideoId:** identifier of the video where the comment was posted
- **PublishedAt:** the date and time when the comment was published
- **TextDisplay:** the original comment text
- **LikeCount:** number of likes received by the comment
- **text_clean:** cleaned version of the comment text used for analysis
- **text_processed:** processed text prepared for text analysis

These attributes allow analysis of user engagement, textual content, and temporal patterns of discussions.

Dataset Size

The dataset contains:

- 3781 comments
- 11 attributes

This dataset size is suitable for performing exploratory analysis and identifying patterns in public discussions.

Exploratory Data Analysis Results

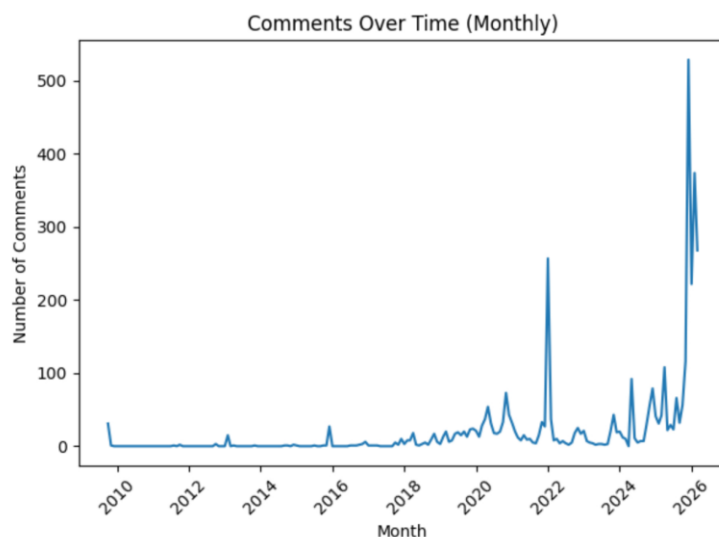


Figure 9: Distribution of YouTube comments over time

This figure shows the number of comments posted on YouTube videos related to flu vaccines over time.

The graph shows that comment activity was relatively low in the earlier years but increased clearly in more recent years. A noticeable increase in the number of comments appears in the later years of the dataset.

This trend may be related to the increase in public discussions about vaccines , especially during times when vaccine topics received more global attention

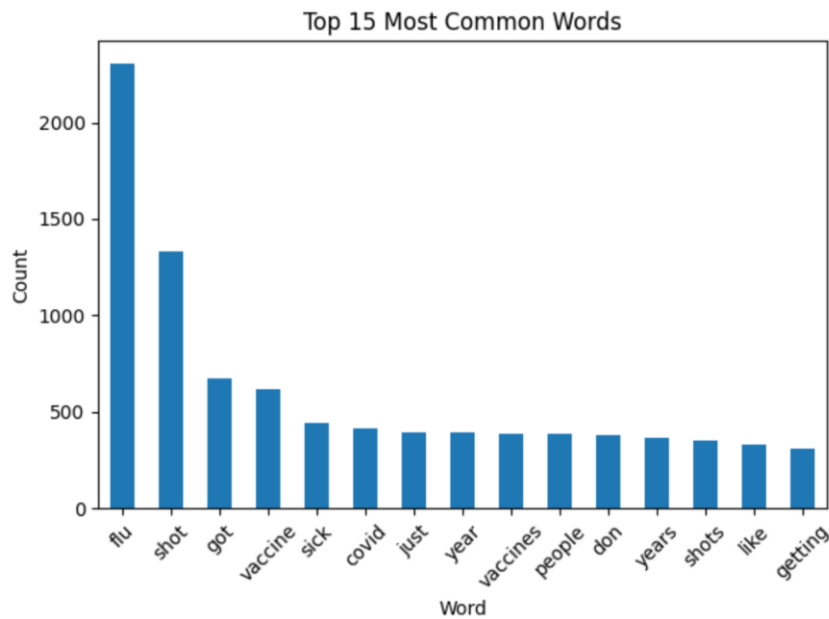


Figure 10: Most common words in YouTube comments

This figure shows the most common words that appear in the YouTube comments. Words such as flu, shot, vaccine, sick, and covid appear among the most common terms. These words show that users often talk about vaccination, illness, and concerns related to vaccines in the comments.

The presence of these words suggests that many users are sharing personal experiences, opinions about vaccines, and other health-related topics

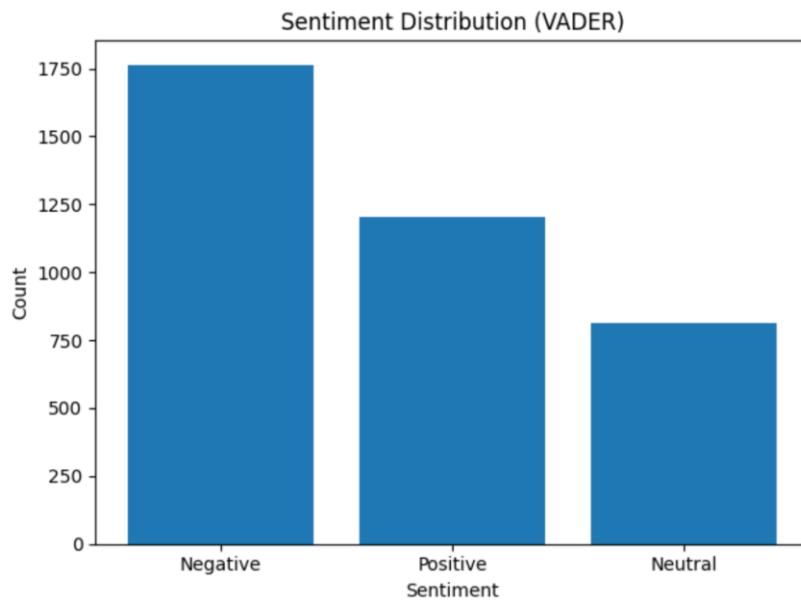


Figure 11: Sentiment distribution of comments

This figure shows the distribution of sentiment in YouTube comments using sentiment analysis.

The results show that negative comments appear more often than positive or neutral comments. Positive comments are also present but in smaller numbers, while neutral comments make up the smallest portion.

This distribution suggests that discussions about vaccines on social media often include concerns, doubts, or criticism, along with some supportive opinions.

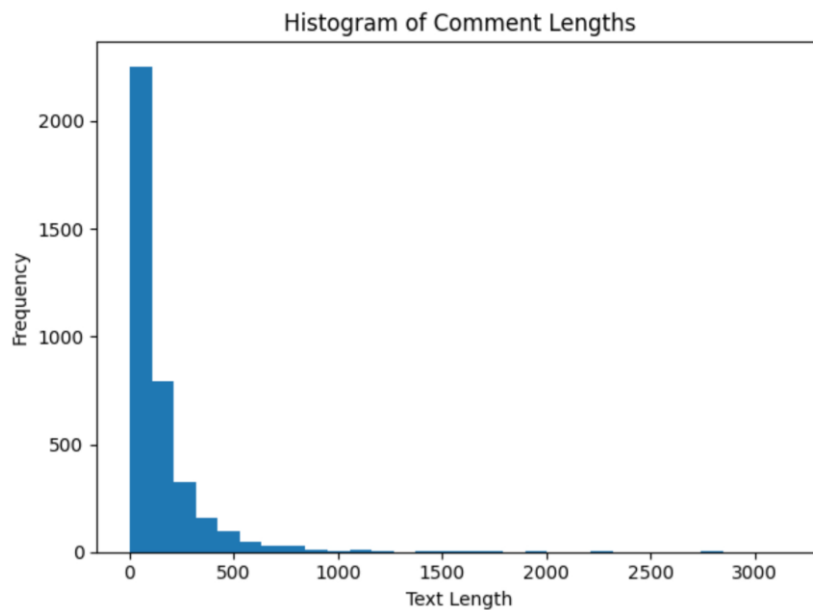


Figure 12: Distribution of comment lengths

This figure shows how article lengths are distributed in the dataset.

Most comments are relatively short , indicating that users often express their opinions briefly. However, a smaller number of comments are significantly longer, suggesting that some users provide detailed explanations or arguments related to vaccine discussions.

This pattern is common in social media datasets where short reactions are more frequent than long discussions.

Summary

The exploratory data analysis of the YouTube dataset provided insights into public discussions about flu vaccines.

The analysis revealed patterns in comment activity over time, common keywords used in discussions , the distribution of sentiment among users, and the characteristics of comment lengths.

Overall, the dataset highlights how social media platforms serve as an important space for public conversations about vaccination.

These insights complement the other datasets used in the project and provide a better understanding of public attitudes toward flu vaccines.

Exploratory Data Analysis (Secondary Data)

Dataset Overview

In addition to the primary datasets used in this project, a secondary dataset was analyzed to provide additional context and support the research findings. The secondary dataset used in this study was downloaded from Kaggle, labeled as “Sentiment towards Vaccination,” where it is publicly available for research purposes.

The dataset originates from the “To Vaccinate or Not to Vaccinate” data science challenge hosted on the Zindi platform, which focuses on sentiment analysis of vaccine-related tweets. Kaggle hosts a copy of this dataset for broader access by researchers and students.

The dataset contains 10,001 tweets related to vaccines and vaccination discussions collected from social media platforms. The purpose of using this dataset is to analyze how people express their opinions about vaccines online and to explore sentiment patterns within these discussions.

Each tweet in the dataset has been labeled with a sentiment category, indicating whether the tweet expresses a positive, neutral, or negative opinion toward vaccines.

This dataset allows researchers to examine patterns in public opinion, textual behavior, and sentiment trends in social media conversations about vaccines.

The dataset contains several important attributes that describe each tweet:

- **tweet_id:** a unique identifier assigned to each tweet. This helps distinguish between tweets and ensures that each record is unique.
- **safe_text:** the cleaned version of the tweet text. Sensitive information such as usernames or links has been removed to ensure privacy and reduce noise in the data.
- **Label:** the sentiment label assigned to the tweet. This column indicates whether the tweet expresses a positive, neutral, or negative sentiment.
- **Agreement:** this column indicates the level of agreement among annotators who labeled the tweet. A higher agreement value means that multiple annotators agreed on the sentiment classification.

These attributes allow the dataset to be used for sentiment analysis and textual analysis, helping researchers understand how opinions about vaccines are expressed on social media.

Exploratory Data Analysis Results

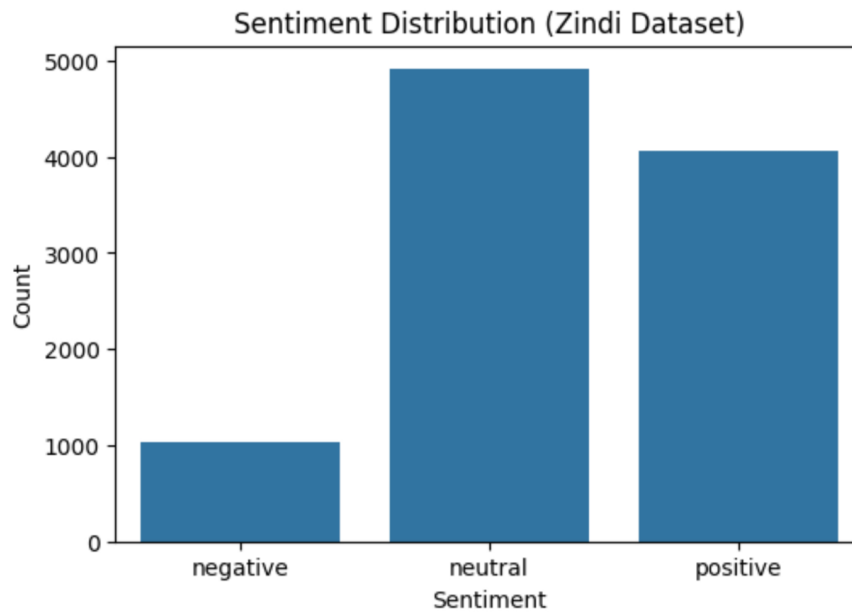


Figure 13: Sentiment Distribution (Zindi Dataset)

From this figure, we can see how sentiment labels were distributed across the dataset.

The dataset contains three sentiment categories which is negative, neutral, and positive.

The results show that neutral tweets represent the largest portion of the dataset, followed by positive tweets , while negative tweets represent the smallest proportion.

This distribution suggests that most discussions about vaccines on social media are neutral or informational rather than strongly negative.

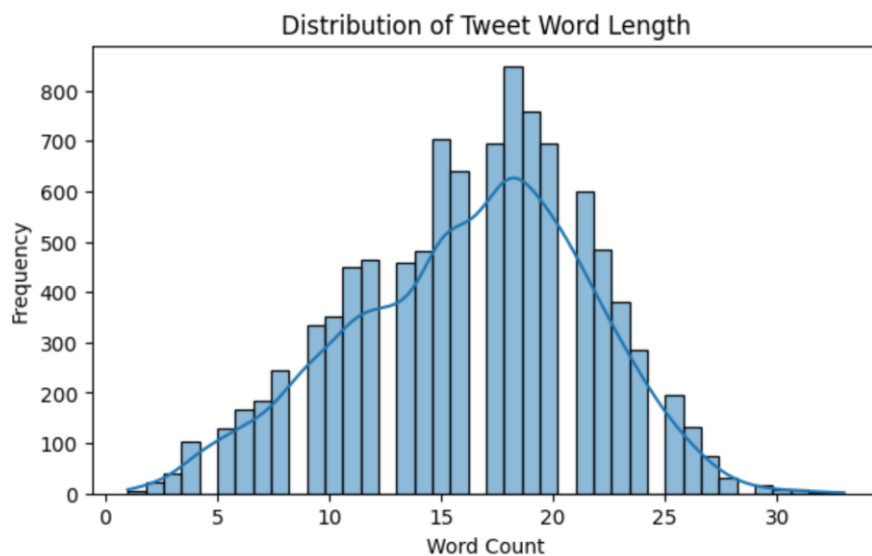


Figure 14: Distribution of Tweet Word Length

This histogram shows the distribution of tweet lengths based on the number of words in each tweet.

The majority of tweets contain approximately 10 to 22 words, with the highest frequency appearing around 16 to 18 words. Only a small number of tweets contain very short or very long texts.

This indicates that tweets in the dataset are generally short and concise, which is consistent with the nature of social media communication.

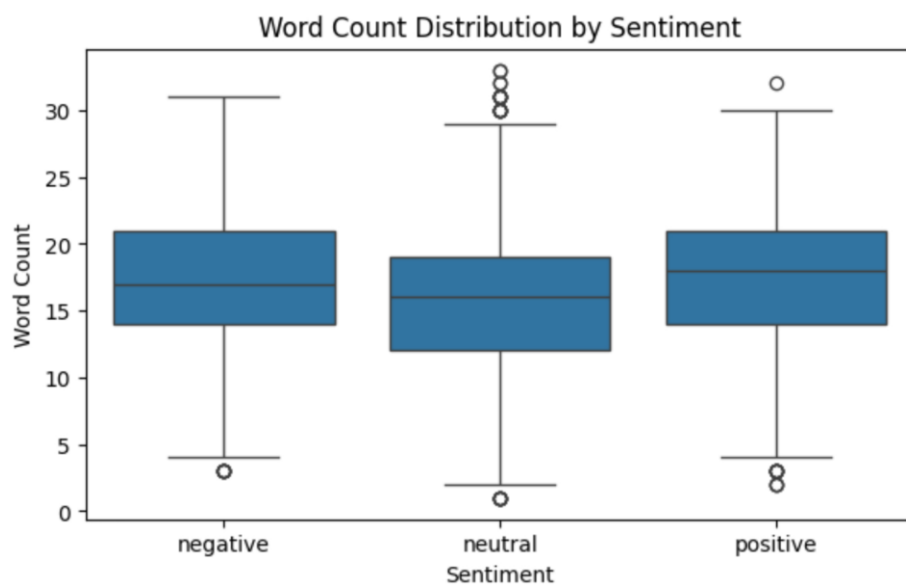


Figure 15: Word Count Distribution by Sentiment

This boxplot compares the distribution of word counts across different sentiment categories: negative, neutral, and positive.

The results show that the word count distribution is relatively similar across all sentiment categories. Most tweets fall within a similar range of word counts, although some outliers with longer texts appear in each category.

This observation suggests that tweet length does not strongly influence the sentiment classification.

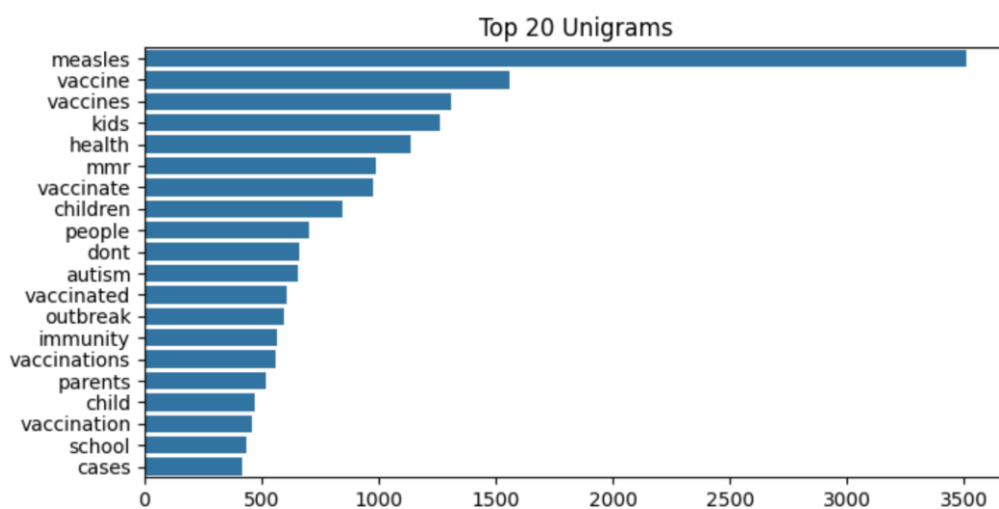


Figure 16: Top 20 Unigrams (Zindi Dataset)

This figure presents the most frequent individual words (unigrams) appearing in the Zindi dataset after text preprocessing.

The results show that vaccine-related terms dominate the discussions, with words such as “measles”, “vaccine”, “vaccines”, and “mmr” appearing most frequently. Other commonly used words include “kids”, “children”, “parents”, and “school”, indicating that many discussions focus on childhood vaccination.

Additionally, terms such as “autism”, “immunity”, and “outbreak” suggest that public conversations often include concerns, misconceptions, or debates surrounding vaccine safety and disease prevention.

Overall , the distribution of these frequent words highlights that the dataset strongly reflects public discussions about vaccination, particularly childhood immunization and related health concerns.

Limitations of the Secondary Data

Although the secondary dataset provides useful information about vaccine related sentiment ,some limitations should be considered when interpreting the results.

First , the dataset was collected from social media platforms, where language is often informal and may include abbreviations , slang, or spelling mistakes. These factors may add noise to the data and affect the accuracy of the text analysis.

Second, the dataset focuses only on vaccine related tweets , which means the results may not represent public opinions about other health topics.

Third, sentiment labels were assigned by human annotators. Even though the dataset includes an agreement score to show how consistent the annotators were, some level of subjectivity may still be present in the labeling process.

Finally, because the dataset was originally collected and prepared by other researchers, the data collection methods and sampling approaches may introduce some bias that cannot be completely controlled in this study.

Comparison Between Primary and Secondary Data

Aspect	GNews Dataset (Primary)	YouTube Dataset (Primary)	OpenFDA Dataset (Primary)	Zindi Dataset (Secondary)
Data Source	News articles collected from online media sources	User comments collected from YouTube videos	Medical adverse event reports from the FDA database	External dataset obtained from a previously published competition on Zindi
Type of Data	Textual news content	Social media user comments	Structured healthcare records	Pre existing structured dataset
Main Objective	Analyze how news articles report vaccine and health related topics	Analyze public opinion and discussions about vaccines in online comments	Analyze reported adverse drug reactions and demographic patterns	Provide additional context or patterns from previously analyzed data
Number of Records	110 news articles	3,781 comments with associated metadata	7,566 medical reaction records	10001 tweets
Key Attributes	url, publishedAt, title, description, content, source_name, text_clean, text_processed, article_length	commentId, videoId, publishedAt, updatedAt, textDisplay, likeCount, text_clean, text_processed	report_id, report_date, patient_age, patient_sex, reaction	Dataset specific attributes depending on the related work
Data Type	Unstructured text + metadata	Unstructured text (user-generated) + engagement metrics	Structured clinical reports	Structured dataset
EDA Focus	Text analysis, keyword frequency, topic trends in news coverage	Sentiment indicators and textual patterns in user comments	Distribution of reactions, age groups, and gender patterns	Observing patterns already studied in related research
Temporal Information	Includes publication time of news articles	Includes timestamps of comments	Includes report dates of medical cases	Depends on the dataset

Insights Generated	News coverage highlights specific health events and vaccine discussions	Public reactions reveal opinions, concerns, and perceptions regarding vaccines	Medical data reveals patterns in reported adverse reactions among different demographic groups	Provides additional supporting patterns for validation
Data Reliability	Comes from news organizations and verified sources	User-generated content, which may contain bias or subjective opinions	Official medical reporting database (FDA)	Secondary source from previous research
Advantages	Provides contextual information about health events and public announcements	Reflects real public sentiment and discussions	Reliable clinical data about drug safety	Enables comparison with findings from other studies
Limitations	News articles may contain editorial bias	Comments may include sarcasm, spam , or emotional reactions	Some demographic fields may contain missing values	Data wasn't collected directly for this project

Statistical Comparison of Key Metrics

Metric	GNews Dataset (Primary)	YouTube Dataset (Primary)	OpenFDA Dataset (Primary)	Zindi Dataset (Secondary)
Dataset Size	110 news articles	3,781 comments	7,566 medical reports	Depends on dataset
Mean (Average)	Mean article length = 518.74 words	Mean likeCount = 4.48	Mean patient age = 56.78	Mean tweet length = 16.27 words
Median	Median article length = 479.5 words	Median likeCount = 0	Median patient age = 61	Median tweet length = 17 words
Main Numerical Variable	article_length	likeCount	patient_age	tweet length
Data Type	Textual news data (converted to numeric article length)	User generated comments	Structured clinical reports	Structured dataset

Key Observations from the Comparison

Several important observations can be identified when comparing the primary and secondary datasets:

1. Different perspectives of the same topic

The primary datasets provide multiple perspectives on the topic of vaccines and health discussions. News articles represent media coverage, YouTube comments represent public opinions, and the OpenFDA dataset represents medical reports from healthcare systems.

2. Differences in data structure

The datasets vary significantly in structure. The GNews and YouTube datasets contain mostly unstructured textual data, which requires preprocessing and natural language processing techniques. In contrast, the OpenFDA dataset contains structured clinical records that allow numerical and demographic analysis.

3. Complementary analytical value

Combining multiple datasets allows a broader understanding of the topic. News data shows how information is communicated to the public, social media data reveals public reactions, and medical data reflects real world clinical outcomes.

4. Role of the secondary dataset

The secondary dataset (Zindi Vaccine Sentiment Dataset) serves as a reference point to compare patterns discovered in the primary datasets. It helps validate whether similar trends appear in other datasets that were previously analyzed.

5. Limitations and data quality considerations

Each dataset has its own limitations. Social media data may contain noise and subjective opinions, while medical datasets may contain missing demographic values. Secondary datasets may also lack detailed documentation about the data collection process.

Contextualizing the Findings

The comparison between the primary datasets (GNews, YouTube, and OpenFDA) and the secondary dataset (Zindi Vaccine Sentiment Dataset) provides broader context for understanding patterns in health related discussions.

The Zindi dataset contains 10,001 tweets labeled with sentiment categories, providing additional insight into how people express their opinions about vaccines on social media.

The GNews dataset reflects how health topics and vaccines are reported in online news media , while the YouTube dataset represents public opinions and reactions expressed in user comments. In contrast, the OpenFDA dataset provides structured medical reports related to drug safety and patient demographics ,offering a clinical perspective on health-related data.

The secondary dataset provides additional support and validation for patterns observed in the primary datasets. By comparing results from multiple data sources, it becomes possible to determine whether similar trends appear across different contexts, such as media coverage, public opinion, and clinical reports.

This multi-source comparison strengthens the reliability of the analysis and provides a more comprehensive understanding of the research topic.

Interpreting Differences Between Datasets

Some differences can appear between the primary and secondary datasets due to variations in data collection methods , data structure , and the nature of the sources.

For example , the GNews and YouTube datasets mainly contain unstructured textual data, which reflects how people discuss or react to health topics in public platforms. In contrast, the OpenFDA dataset contains structured clinical reports collected from healthcare systems, which focus on medical records rather than public discussions.

If differences are observed between patterns in the primary datasets and those in the secondary dataset , these variations may be explained by factors such as differences in sampling methods, time periods of data collection, or geographical coverage. Additionally, social media data may include subjective opinions or noisy content, while clinical datasets may contain missing demographic information.

Therefore, comparing multiple datasets helps identify both consistent patterns and possible discrepancies in the analysis.

Summary of New Insights and Hypotheses

The exploratory data analysis (EDA) conducted on the primary and secondary datasets provided several important insights into the characteristics of health related discussions and medical reports.

First , the analysis of the GNews dataset showed that news articles vary in length but generally follow a consistent distribution, with a median article length of approximately 480 words. This suggests that most news articles provide moderate levels of detail when reporting health related topics.

Second, the YouTube dataset revealed that user engagement with comments is generally low. The median value of the likeCount variable is zero, indicating that most comments receive no likes , while a small number of comments receive significantly higher engagement. This indicates that public discussions on social media often contain many low interaction comments and a few highly visible ones.

Third , the OpenFDA dataset provided structured clinical data that allowed the analysis of demographic patterns. The mean patient age was approximately 56.78 years, and the median age was around 61 years. This suggests that many reported adverse drug reactions occur among middle aged and older patients.

By comparing the primary datasets with the secondary dataset (Zindi Vaccine Sentiment Dataset), which contains 10,001 labeled tweets, the analysis helped provide additional context for interpreting the patterns observed in the data. The secondary dataset helps support and validate trends identified in the primary sources and demonstrates how insights can be strengthened by combining multiple datasets from different domains.

Based on these observations, several hypotheses can be proposed. One possible hypothesis is that public reactions on social media may not always reflect clinical medical trends observed in healthcare datasets. Another hypothesis is that media coverage , public opinion, and clinical reports each represent different perspectives on the same health topics , and combining these sources can provide a better understanding of the issue.

Machine Learning Models

In this phase of the project, we applied different machine learning models to classify sentiment in text data related to vaccines. The main goal was to see how well each model can determine if the text is positive, neutral, or negative.

To do this, we combined labeled data from the GNews and YouTube datasets and prepared it for training. We then used TF-IDF to convert the text into numerical features that the models can understand.

Three models were used in this phase: Logistic Regression as a baseline, Support Vector Machine (SVM) , and Random Forest. Each model was trained and tested, then evaluated based on its accuracy and performance on different sentiment classes.

In the following sections , we present the results of each model and compare their performance.

Baseline Model (Logistic Regression)

The baseline model was developed using Logistic Regression combined with TF-IDF vectorization to convert textual data into numerical features suitable for machine learning.

For this model, labeled data from both the GNews and YouTube datasets were combined to create a unified dataset for sentiment classification. The text data was first preprocessed using the same cleaning steps applied in earlier phases, including lowercasing, removing noise, and generating cleaned text columns. After preprocessing, TF-IDF was applied with a maximum of 5000 features and English stopword removal to represent the text data numerically.

The dataset was then split into training and testing sets using an 80/20 ratio. To handle class imbalance, the Logistic Regression model was configured with class balancing.

The baseline model achieved an overall accuracy of approximately 67%, which provides a reasonable starting point for evaluating model performance.

In terms of class-wise performance:

- Negative (-1): F1-score = 0.75
- Neutral (0): F1-score = 0.59
- Positive (1): F1-score = 0.57

The model shows strong performance in detecting negative sentiment, indicating that negative expressions are easier to identify in the dataset. However, the performance for neutral and positive classes is lower, which may be due to overlapping language patterns and class imbalance.

Overall, the baseline model provides a useful benchmark for comparison. While its performance is acceptable, there is clear room for improvement, particularly in distinguishing between neutral and positive sentiments. This justifies the need to explore more advanced models in the next steps.

Support Vector Machine (SVM) Mode

To improve the results, a Support Vector Machine (SVM) model was used. This model is known to work well with text data because it can separate between different classes more effectively.

The same dataset used in the baseline model was used here, which combines labeled data from GNews and YouTube. The text was already cleaned in the preprocessing step, and TF-IDF was used again to convert the text into numerical features.

In this model , TF-IDF was slightly improved by using both single words (unigrams) and pairs of words (bigrams). This helps the model understand the context of the text better instead of looking at each word alone. The dataset was split into training and testing sets using the same 80/20 ratio.

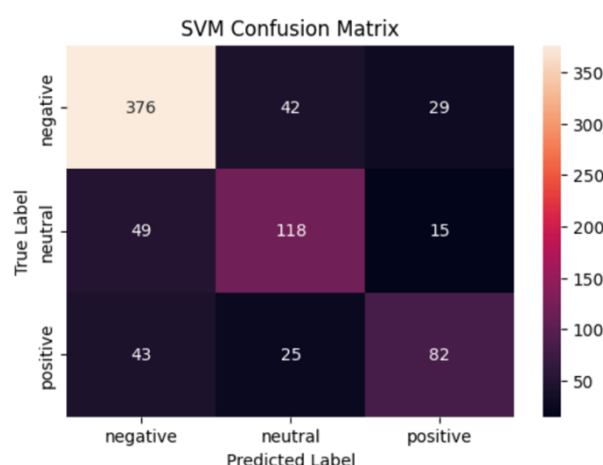
The SVM model achieved an accuracy of approximately 74%, which is higher than the baseline model.

For class performance:

- Negative (-1): F1-score ≈ 0.82
- Neutral (0): F1-score ≈ 0.64
- Positive (1): F1-score ≈ 0.59

The results show that the SVM model improved the performance, especially in detecting negative sentiment. It also showed better results for neutral and positive classes compared to the baseline model.

Overall , this model performed better than Logistic Regression because it can handle text data more effectively and capture patterns in the data more clearly but there is still some difficulty in distinguishing between neutral and positive sentiments.Figure



This figure shows the confusion matrix of the SVM model. The model performs well in predicting negative sentiment, while there is some confusion between neutral and positive classes, indicating that these categories are more difficult to distinguish.

Figure 17: SVM Confusion Matrix

Random Forest Model

To further compare the performance, a Random Forest model was used. This model is based on multiple decision trees and is useful for capturing more complex patterns in the data.

The same dataset was used again, combining labeled data from GNews and YouTube. The text was already cleaned, and TF-IDF was applied to convert the text into numerical features. Similar to the SVM model, both unigrams and bigrams were used to improve the representation of the text.

The dataset was split into training and testing sets using the same 80/20 ratio.

The Random Forest model achieved an accuracy of approximately 72%.

For class performance:

- Negative (-1): F1-score $\approx$ 0.81
- Neutral (0): F1-score $\approx$ 0.63
- Positive (1): F1-score $\approx$ 0.45

The results show that the model performed well in detecting negative sentiment, but it had difficulty identifying positive sentiment compared to the other models.

Overall, the Random Forest model gave good results, but it was slightly lower than the SVM model in terms of overall performance. This suggests that while Random Forest can capture complex patterns, it may not be as effective as SVM for this type of text classification task.

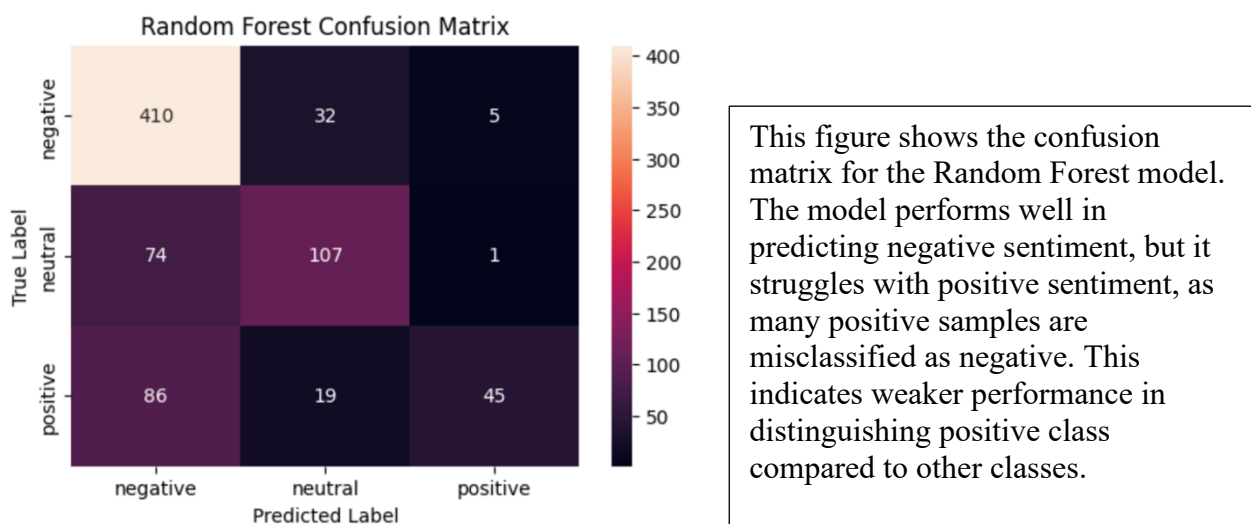


Figure 18: Random Forest Confusion Matrix

Model Comparison

After training and evaluating all three models on the same dataset and test set, the results were compared based on accuracy, precision, recall, and F1-score. The table below summarizes the performance of each model:

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.7112	0.7197	0.7112	0.7142
SVM	0.7227	0.7187	0.7227	0.7201
Random Forest	0.6945	0.7003	0.6945	0.6719

The comparison shows that SVM achieved the highest accuracy and F1-score among all models, followed by Logistic Regression, while Random Forest showed the lowest overall performance. All models struggled with neutral and positive classes due to class imbalance in the dataset.

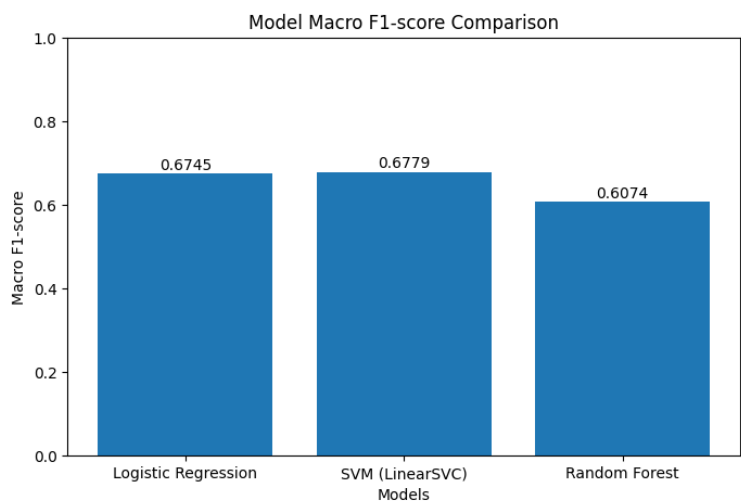


Figure 19: Model Macro F1-score Comparison

Since the dataset is imbalanced, accuracy alone is not a reliable metric. Therefore, we used the macro F1-score to evaluate model performance, as it considers both precision and recall while giving equal importance to all classes. Based on this metric, the SVM model achieved the highest performance and was selected as the best model.

Final Model Selection

Based on the evaluation results, the SVM model was selected as the best performing model for sentiment classification. It achieved the highest accuracy of 72.27% and the highest weighted F1-score of 0.7201. Compared to Logistic Regression and Random Forest, SVM demonstrated more balanced performance across all three sentiment classes. Additionally, SVM is well-suited for text classification tasks that rely on high-dimensional TF-IDF features, as it effectively finds the optimal decision boundary between classes. Therefore, SVM was chosen as the final model for predicting sentiment in vaccine-related comments and news articles.

Model Predictions

The selected model was used to predict sentiment labels (negative, neutral, and positive) on unseen test data.

The results show that the model is more accurate in identifying negative sentiment, while neutral and positive sentiments are slightly more difficult to classify. This is common in text data where some expressions may overlap between categories.

Conclusion

In this project, we analyzed public opinions about flu vaccines using data from different sources such as GNews, YouTube, and OpenFDA. Each dataset provided a different perspective, which helped us better understand how vaccines are discussed in both media and public platforms.

After applying data preprocessing and exploratory analysis, we trained three machine learning models: Logistic Regression, SVM, and Random Forest. The results showed that the SVM model achieved the best overall performance compared to the other models.

However, the evaluation was not based only on accuracy. Since the dataset is imbalanced, accuracy alone can be misleading. For this reason, we also considered other metrics such as F1-score, precision, and recall for each class. These metrics provided a more realistic understanding of model performance across negative, neutral, and positive sentiments.

The results showed that all models performed better in detecting negative sentiments, while neutral and positive sentiments were more difficult to classify. This reflects the complexity of text data and the overlap between different sentiment categories.

In conclusion, this project highlights the importance of using multiple evaluation metrics and combining different data sources to achieve more reliable results. It also shows that SVM is the most suitable model for this task based on its balanced performance.

References

Topic	Source
Study 1	Examining the Negative Sentiments Related to Influenza Vaccination from 2017 to 2022: An Unsupervised Deep Learning Analysis of 261,613 Twitter Posts - PMC
Study 2	Examining Public Messaging on Influenza Vaccine over Social Media: Unsupervised Deep Learning of 235,261 Twitter Posts from 2017 to 2023 MDPI
Study 3	https://pmc.ncbi.nlm.nih.gov/articles/PMC9014796/
Study 4	Understanding influenza vaccination attitudes and behaviors: An assessment of health decision-making preferences - ScienceDirect
Study 5	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0129993
Secondary Data	https://www.kaggle.com/datasets/alfredkondoro/to-vaccinate-or-not-to-vaccinate-zindi?resource=download&select=Train.csv